

Advanced Image Processing

Assignment 3

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Problem 1: Block Matching and 3D Filtering (BM3D)

Assumed noise variance of the noisy image: 0.0625 The true noise variance of the noisy image is:

$$\hat{\sigma}_Z^2 = \text{MSE}(\text{clean image, noisy image}) = \mathbf{0.025800}$$

Q1.1. Comparing MSE Performance: Stage 1 vs Stage 2

The BM3D algorithm runs in two stages:

- **Stage 1 (Hard Thresholding):** The blocks are grouped and transformed, and then a hard-thresholding operation is applied. This produces a basic denoised estimate of the clean image.
- **Stage 2 (Wiener Filtering):** The Stage 1 output is then used as a pilot to guide Wiener filtering on the noisy image, which produces a refined estimate of the clean image.

Results:

Stage	Method	MSE
Stage 1	Hard Thresholding only	0.012854
Stage 2	Hard Thresholding + Wiener Filtering	0.009960

Observation: Stage 2 achieves a lower MSE (0.009960) compared to Stage 1 (0.012854). In stage 2, the Wiener filter uses the denoised image of Stage 1 as a pilot to compute better spectral coefficients, which reduces the residual noise more effectively.

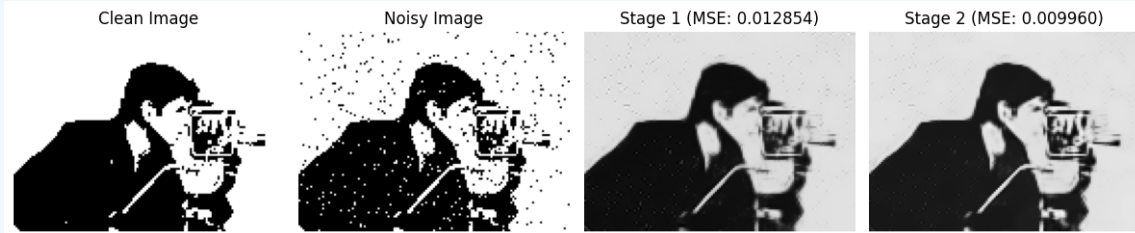


Figure 1: Visual comparison: Clean Image, Noisy Image, Stage 1 denoised image, Stage 2 denoised image.

Q1.2. MSE vs Noise Variance σ_Z^2

Experiment: The BM3D algorithm was run with multiple noise variance σ_Z^2 over the range from 0 to 1.0. The MSE between the denoised image and the clean image was recorded at each input noise variance.

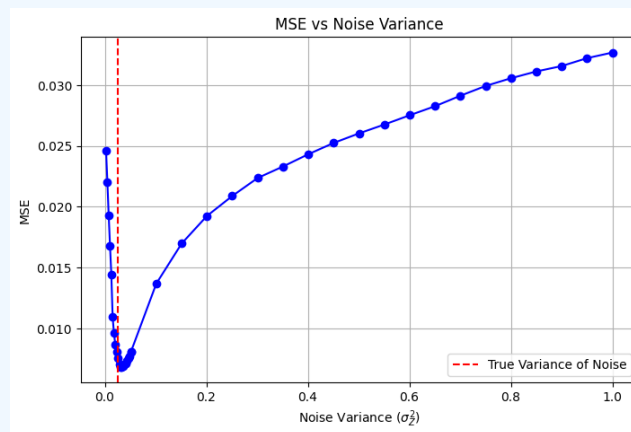


Figure 2: MSE vs. input noise variance σ_Z^2 . The red dashed line marks the true noise variance.

Observations:

- **Minima:** The curve has a minima around the true noise variance (marked by the red dashed vertical line).
- **Under-estimation** ($\sigma_Z^2 < \hat{\sigma}_Z^2$): If the input noise variance is too low, the algorithm underestimates the noise and does not remove it fully, which leads to a high MSE.
- **Over-estimation** ($\sigma_Z^2 > \hat{\sigma}_Z^2$): If the input noise variance is too high, the algorithm overestimates the noise. Because of this it blurs out the fine details and which increasing MSE.

Q1.3. Replacing Wiener Filter with Hard Thresholding in Stage 2

Methodology: To replace the Wiener filter in Stage 2 with hard thresholding, the following steps were followed:

1. Run Stage 1 (Hard Thresholding) to obtain the pilot image.
2. Extract the block-match coordinates from the pilot image.
3. Apply Hard Thresholding to the noisy image using these block-match coordinates.

Results:

Stage 2 Method	MSE
Wiener Filtering	0.009960
Hard Thresholding	0.012855

Observations: Wiener Filtering performs better (MSE = 0.009960) than Hard Thresholding (MSE = 0.012855) in the 2nd stage. Wiener coefficients are optimal in the least-squares sense for signal reconstruction, whereas Hard Thresholding just discards small coefficients.

Problem 2: Zero Reference Deep Curve Estimation

Hyperparameters: $\text{lr} = 10^{-5}$, epochs = 50, mean_val = 0.40, tv_weight = 100, col_weight = 4, exp_weight=10.

PSNR and SSIM Comparisons (LOL Dataset):

Image ID	Low-Light Input		Pre-Trained		Fine-Tuned	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
00692	7.80	0.1111	13.83	0.6266	19.09	0.6962
00702	5.63	0.1003	10.00	0.4873	12.59	0.5378
00719	10.38	0.4387	16.77	0.8178	16.14	0.8231
00782	8.63	0.1656	16.30	0.7736	20.22	0.7930
00743	10.48	0.3095	23.07	0.7931	22.86	0.8001

Observations:

- For all the test images, fine-tuning improves the SSIM value, indicating that image-specific fine-tuning is beneficial.
- For images **00692**, **00702**, and **00782**, fine-tuning yields substantially higher PSNR and SSIM.
- For **00719** and **00743**, the pretrained model achieves slightly higher PSNR, but fine-tuning achieves a slightly higher SSIM.
- Overall, fine-tuning improves or maintains performance across all images.

Visual Comparisons (LOL Dataset)



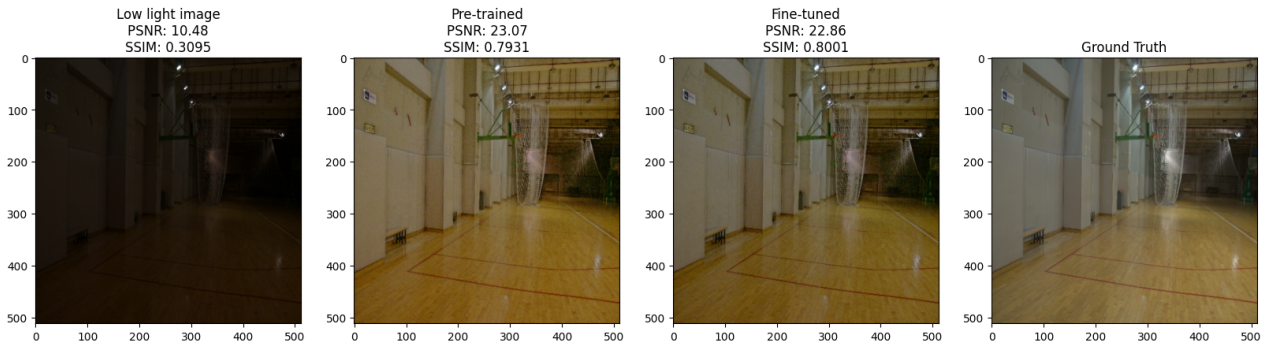


Figure 3: Visual Comparison of LOL Dataset Images: Low-light input, Pre-trained and Fine-tuned Zero-DCE outputs, Ground Truth.

Visual Comparisons (Smartphone Captured Low-Light Images)

